MONTH 1: INTRODUCTION TO DATA SCIENCE & MACHINE LEARNING

- Week 1: Data Science Basics
- Tasks:
  - Introduction to data science concepts, data types, and basic statistics.
  - Data preprocessing: handling missing values, normalization.
- Project: Perform exploratory data analysis (EDA) on a dataset (e.g., Titanic dataset).
- Week 2: Supervised Learning I
- Tasks:
  - Learn linear regression and logistic regression.
  - Understand overfitting, underfitting, and model evaluation (cross-validation).
- Project: Build a linear regression model to predict housing prices.
- Week 3: Supervised Learning II
- Tasks:
  - Explore classification algorithms like Decision Trees, Random Forests.
- Project: Implement a classification model to predict customer churn.
- Week 4: Unsupervised Learning
- Tasks:
  - Learn about clustering algorithms (K-Means, Hierarchical Clustering).
- Project: Apply clustering to segment customers based on purchasing behavior.

MONTH 2: ADVANCED MACHINE LEARNING TECHNIQUES

- Week 5: Feature Engineering & Selection
- Tasks:
  - Learn about feature engineering techniques and feature selection methods.
- Project: Improve a previous model by applying advanced feature engineering.
- Week 6: Neural Networks Basics
- Tasks:
  - Introduction to neural networks, activation functions, and backpropagation.
- Project: Implement a simple neural network from scratch to classify handwritten digits (MNIST dataset).
- Week 7: Deep Learning (CNNs)
- Tasks:
  - Learn about Convolutional Neural Networks (CNNs) for image processing.
- Project: Build a CNN model for image classification (e.g., CIFAR-10 dataset).
- Week 8: Natural Language Processing (NLP)
- Tasks:
  - Understand basic NLP concepts: text preprocessing, word embeddings (Word2Vec).
- Project: Develop a sentiment analysis model for Twitter data.

MONTH 3: REAL-WORLD APPLICATIONS AND DEPLOYMENT

- Week 9: Model Deployment
- Tasks:
  - Learn to deploy machine learning models using Flask/Django.
- Project: Deploy a trained model as a web service using Flask.
- Week 10: Time Series Analysis
- Tasks:
  - Introduction to time series forecasting techniques (ARIMA, LSTM).
- Project: Forecast stock prices using time series analysis.
- Week 11: Model Optimization
- Tasks:
  - Learn about model optimization (grid search, random search).
- Project: Optimize a previous model and compare the performance before and after optimization.
- Week 12: Final Project & Presentation
- Project: Develop a final project that integrates various aspects of machine learning and data science (e.g., end-to-end ML pipeline). Present the project with a live demo and documentation.